

A Randomised Controlled Trial on the Impact of Parenting on Adolescent Emotion Regulation-Related Brain Function and Mental Health

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2026 flux Congress



A word cloud featuring various emotions and feelings. The words are arranged in a dense, overlapping manner, with some words being significantly larger than others. The colors of the words range from light blue to dark purple, with some words in white. The background is a solid dark color.

Gratitude Uncertainty Disquietude
Shame Restlessness Emptiness
Disorientation Consternation Fear
Dismay Patience Terror
Untetheredness Wonder Betrayal
Disappointment Joy Confusion
Rage Inertia Calm Self-Loathing
Worry Pride Melancholy Angst
Anger Overwhelm Saudade
Freneticism Creativity Envy Lethargy
Loneliness Ambivalence Frustration
Inspiration Bliss Sorrow
Befuddlement Relief Awe Yearning Fretfulness Nostalgia
Despair Disillusionment Vulnerability Bewilderment
Shock Delight Helplessness Euphoria
Hope Sadness Malaise
Grief Apprehension

Family influence on emotion regulation development



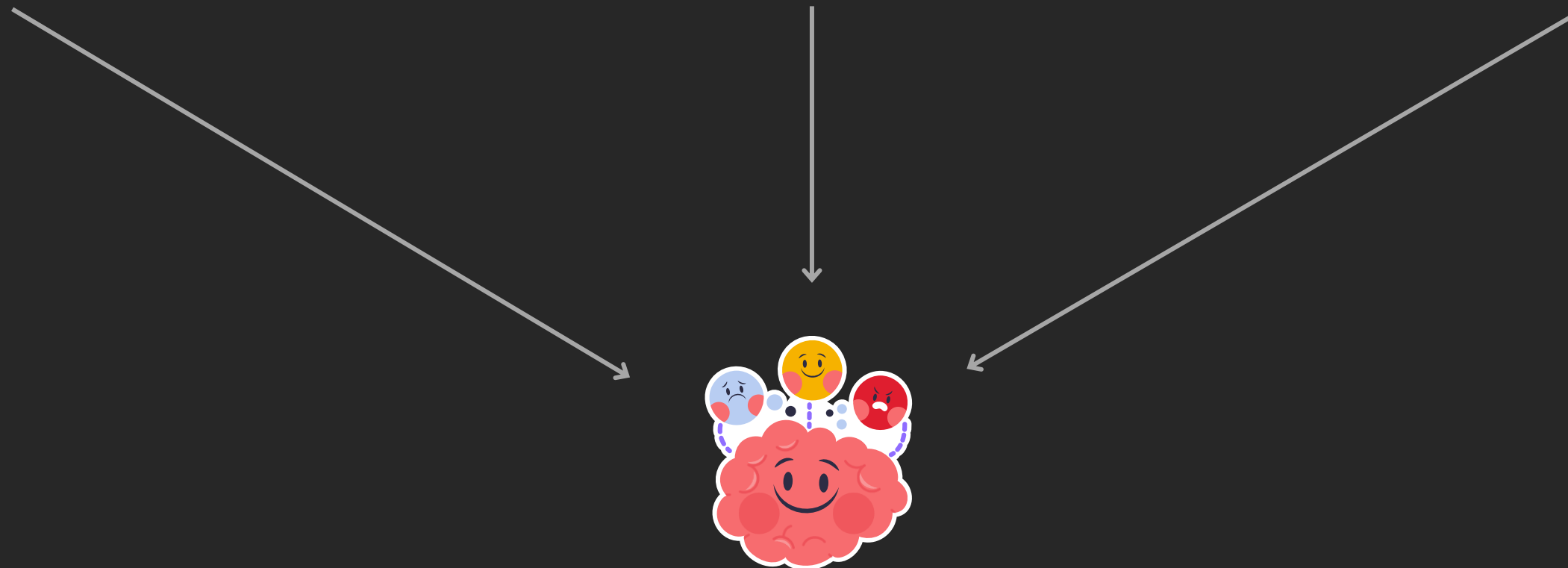
Role modelling



Response to emotions



Family emotional climate



Morris et al., 2007; 2017
Lin et al., 2024

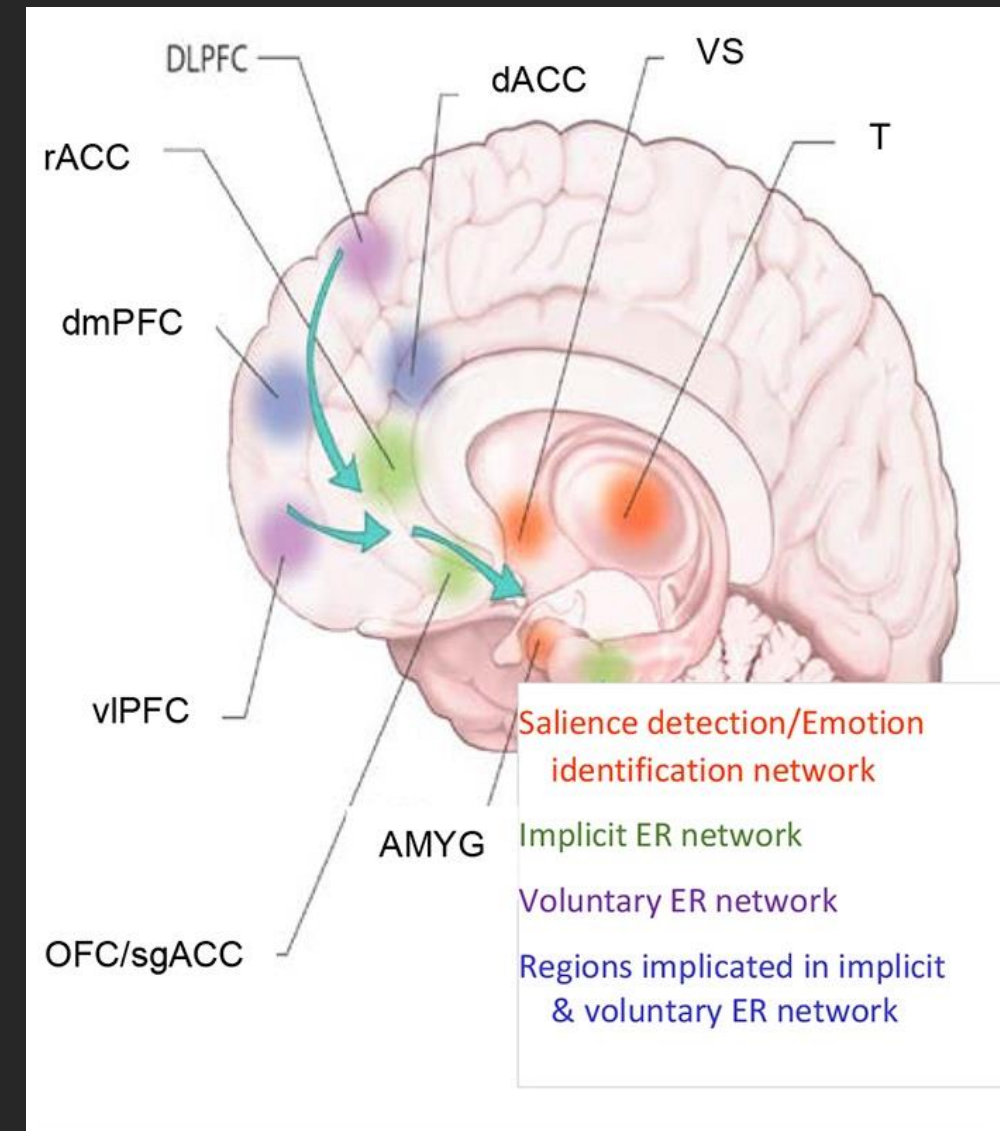
Review > Dev Psychol. 2020 Mar;56(3):516-527. doi: 10.1037/dev0000893.

A review of associations between parental emotion socialization behaviors and the neural substrates of emotional reactivity and regulation in youth

Patricia Z Tan¹, Caroline W Oppenheimer², Cecile D Ladouceur², Rosalind D Butterfield³, Jennifer S Silk³

Parental Influences on Neural Mechanisms Underlying Emotion Regulation

[Kara L Kerr](#)^{a,c}, [Erin L Ratliff](#)^{a,c}, [Kelly T Cosgrove](#)^{b,c}, [Jerzy Bodurka](#)^{c,d}, [Amanda Sheffield Morris](#)^{a,c}, [W Kyle Simmons](#)^{b,c,e}



**Can we make causal
inferences?**

Early Parenting Intervention Effects on Brain Responses to Maternal Cues Among High-Risk Children

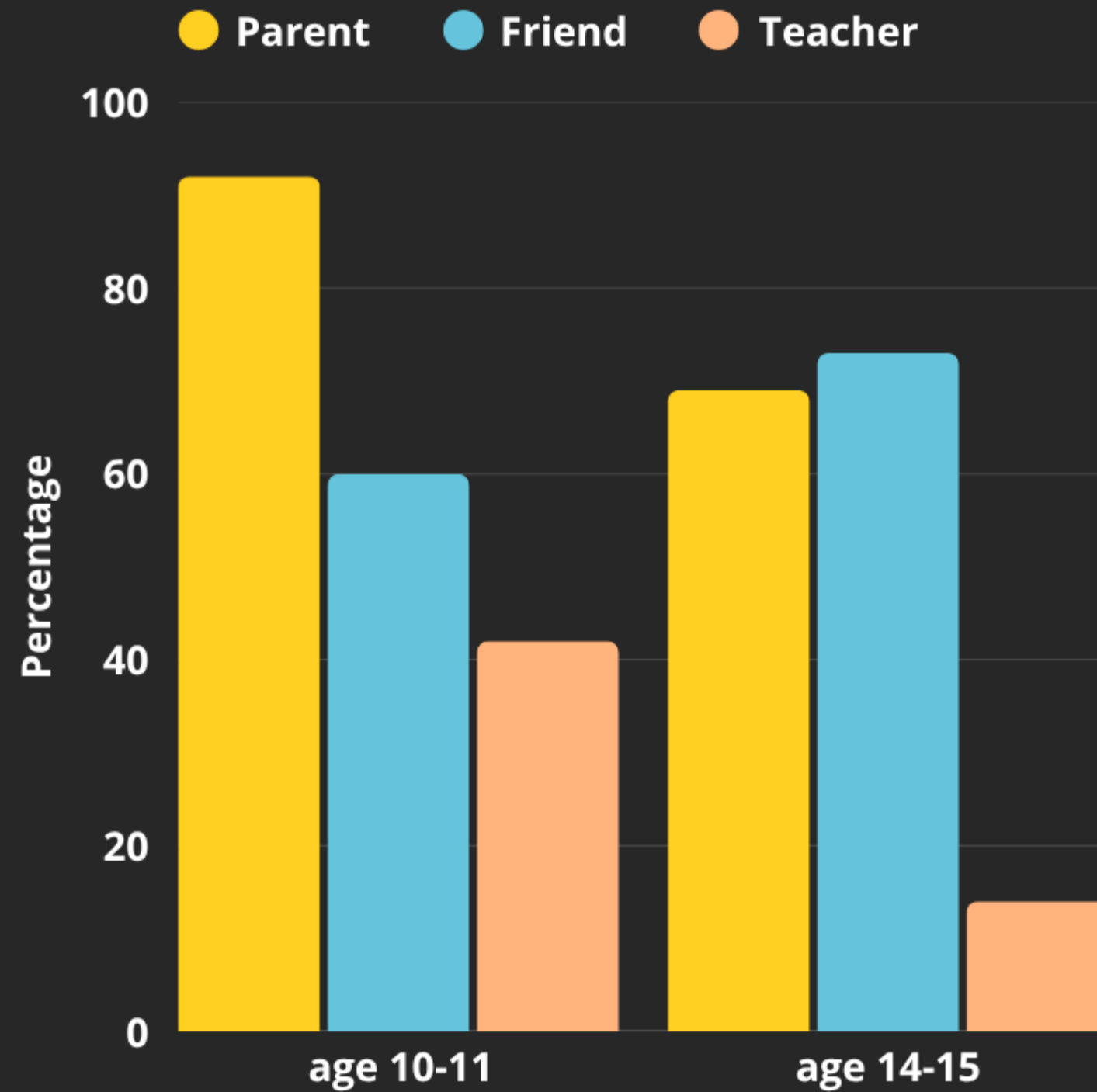
Emilio A. Valadez, Ph.D., Nim Tottenham, Ph.D., Alexandra R. Tabachnick, M.S., Mary Dozier, Ph.D.

Preliminary examination of the effects of an early parenting intervention on amygdala-orbitofrontal cortex resting-state functional connectivity among high-risk children: A randomized clinical trial

Marta Korom¹ , Emilio A. Valadez², Nim Tottenham³ , Mary Dozier¹  and Jeffrey M. Spielberg¹

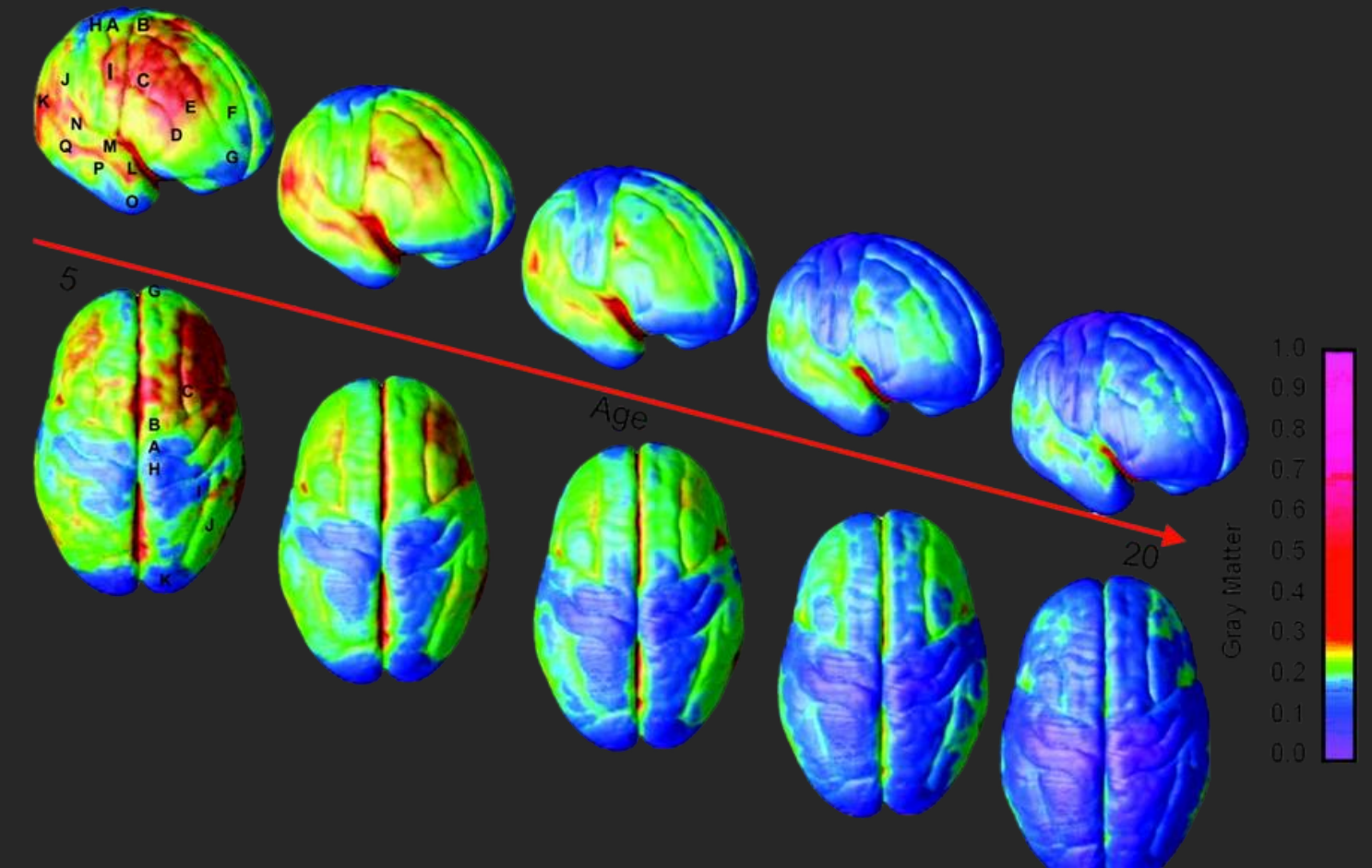
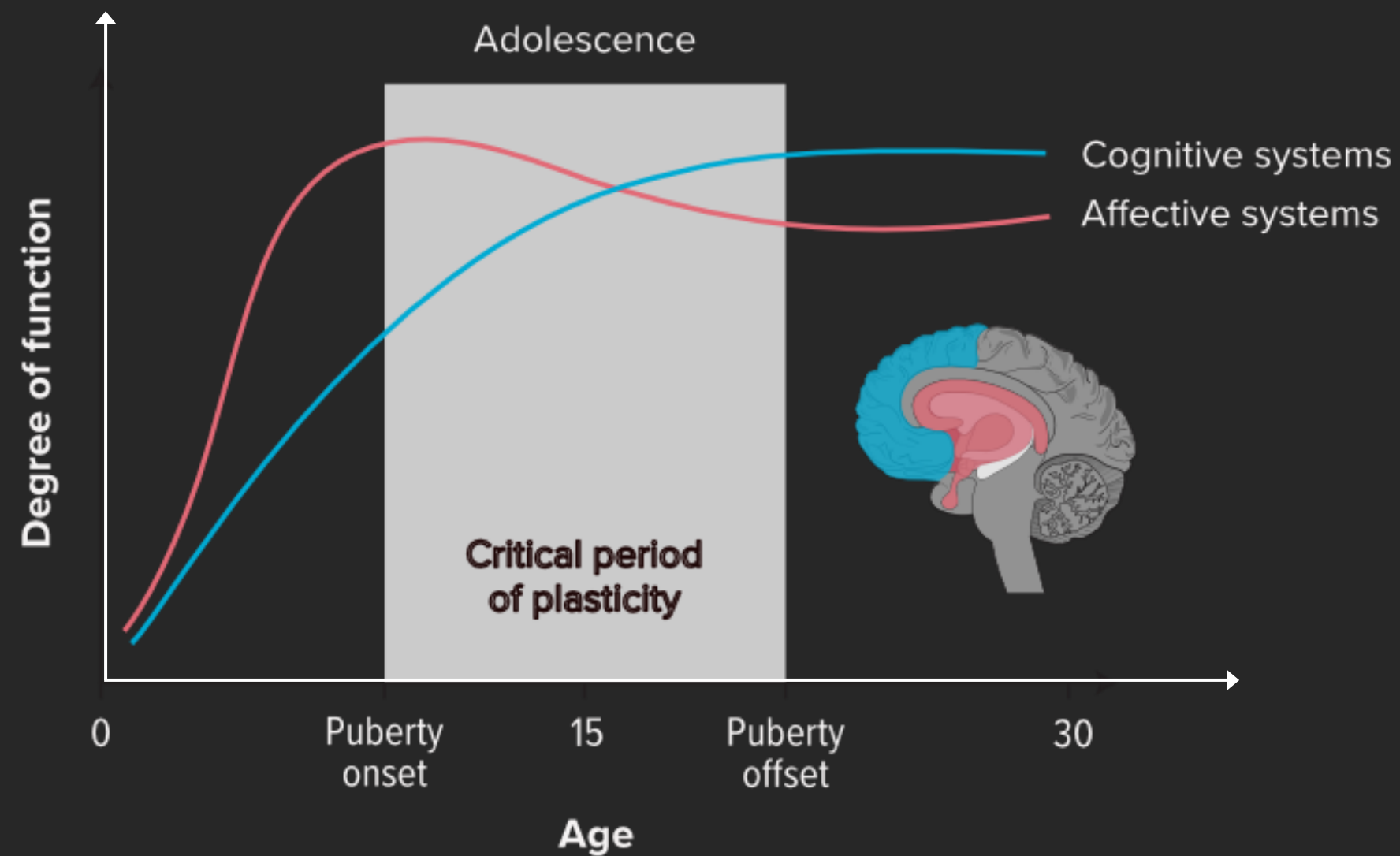
¹Department of Psychological and Brain Sciences, University of Delaware, Newark, DE, USA, ²Department of Human Development and Quantitative Methodology, University of Maryland, College Park, MD, USA and ³Department of Psychology, Columbia University, New York, NY, USA



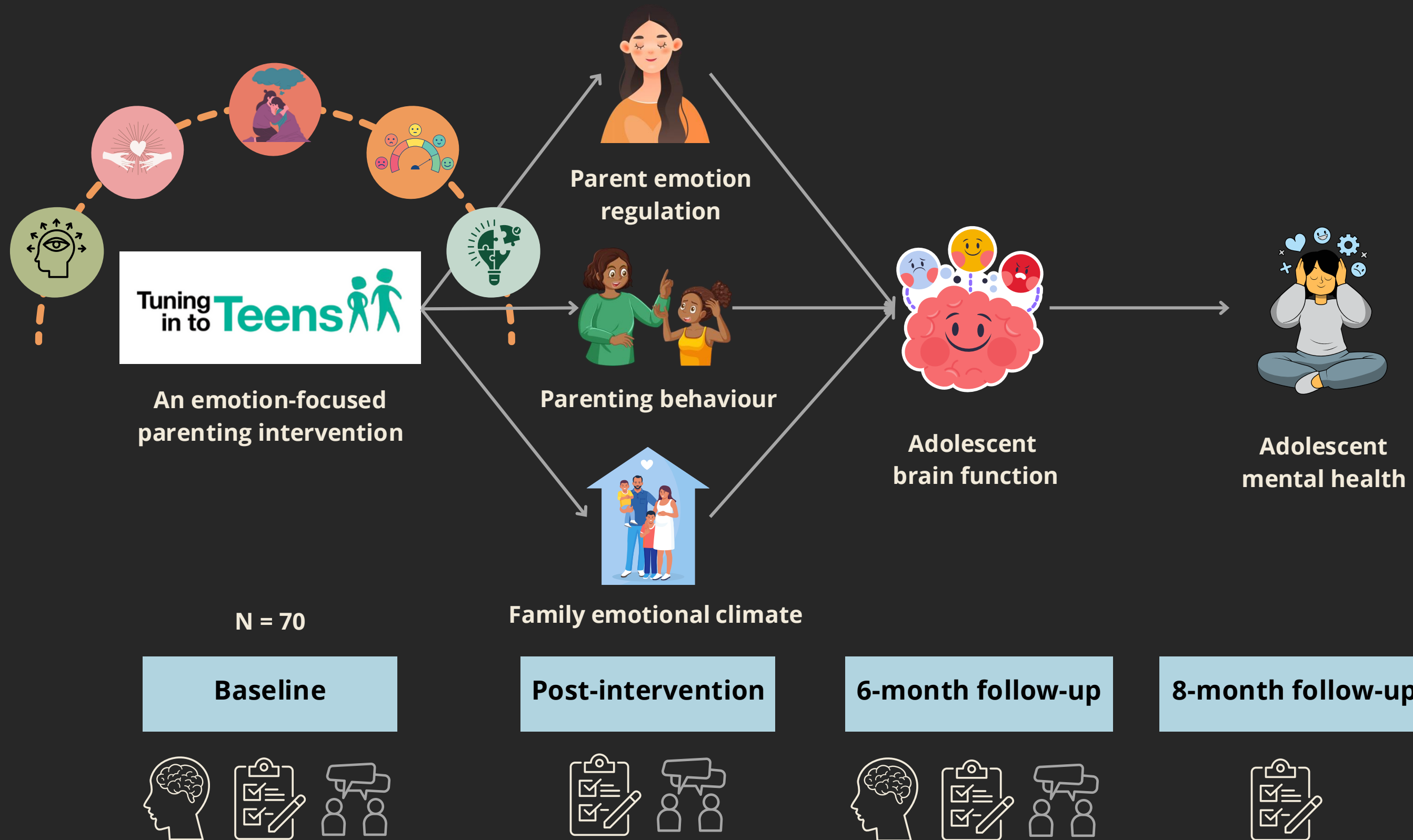


**Who do you go for help
for a personal or
emotional problem?**

Early adolescence is the **most** dynamic period of brain development, second only to infancy



Gogtay et al., 2004; Mills et al., 2016;
Powers & Casey, 2015; Somerville et al., 2010



73%

Adolescents are
White/Caucasian

11.45

Mean adolescent age

84%

Mothers have a
Bachelor's degree or
higher

40%

Mothers currently
receiving a mental
health treatment

61.33

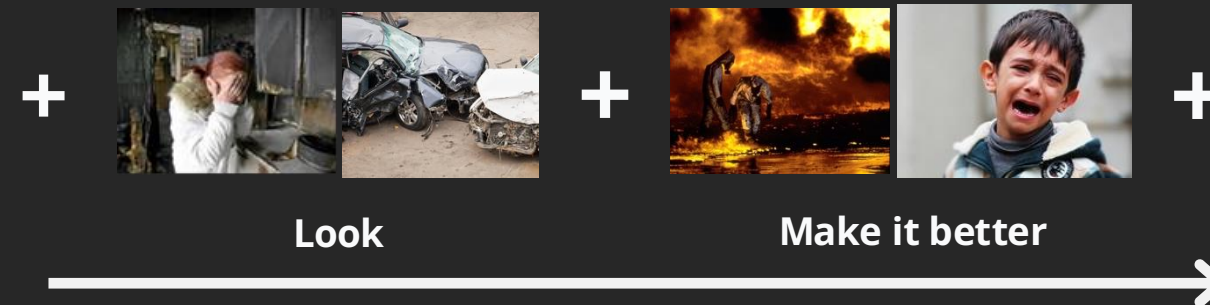
Mean RCADS score
(>44 indicating elevated
depressive/anxiety symptoms)

73%

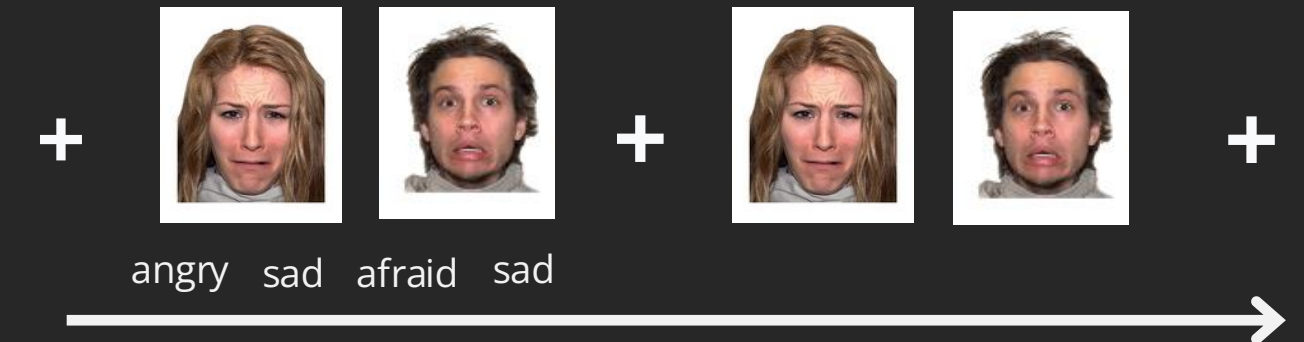
Family annual income
> AUD 100k



Explicit emotion regulation (Reappraisal > Look)

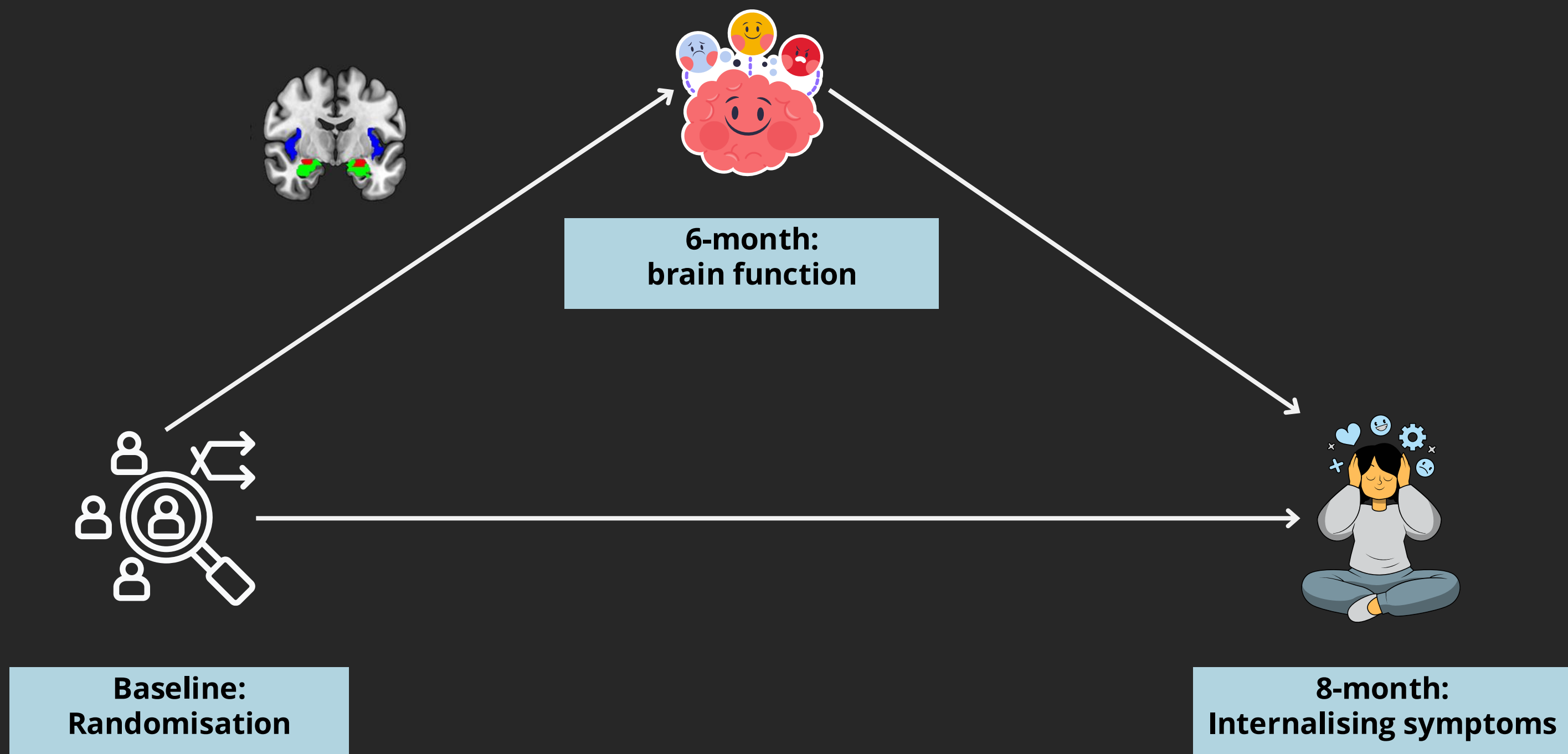


Implicit emotion regulation (Affect label > Observe)

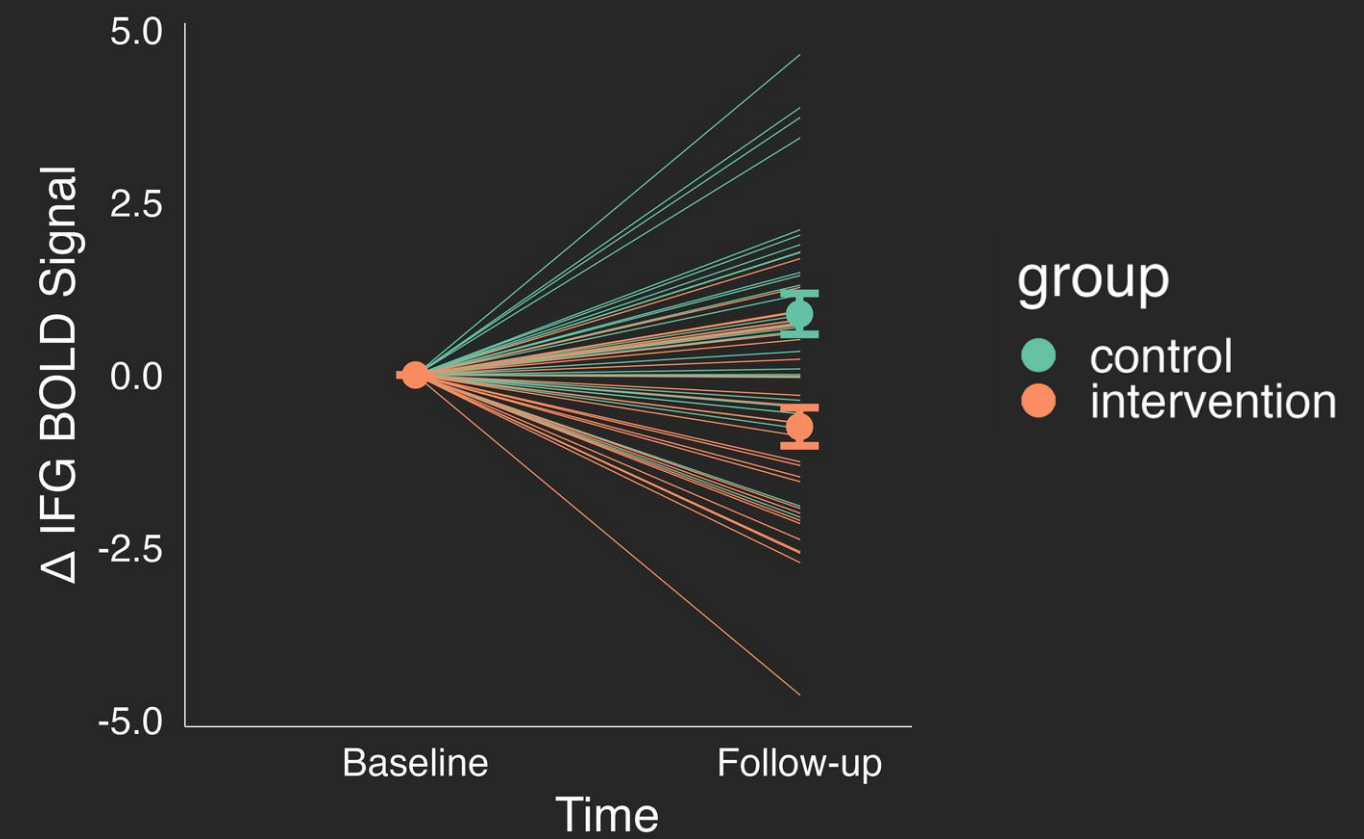
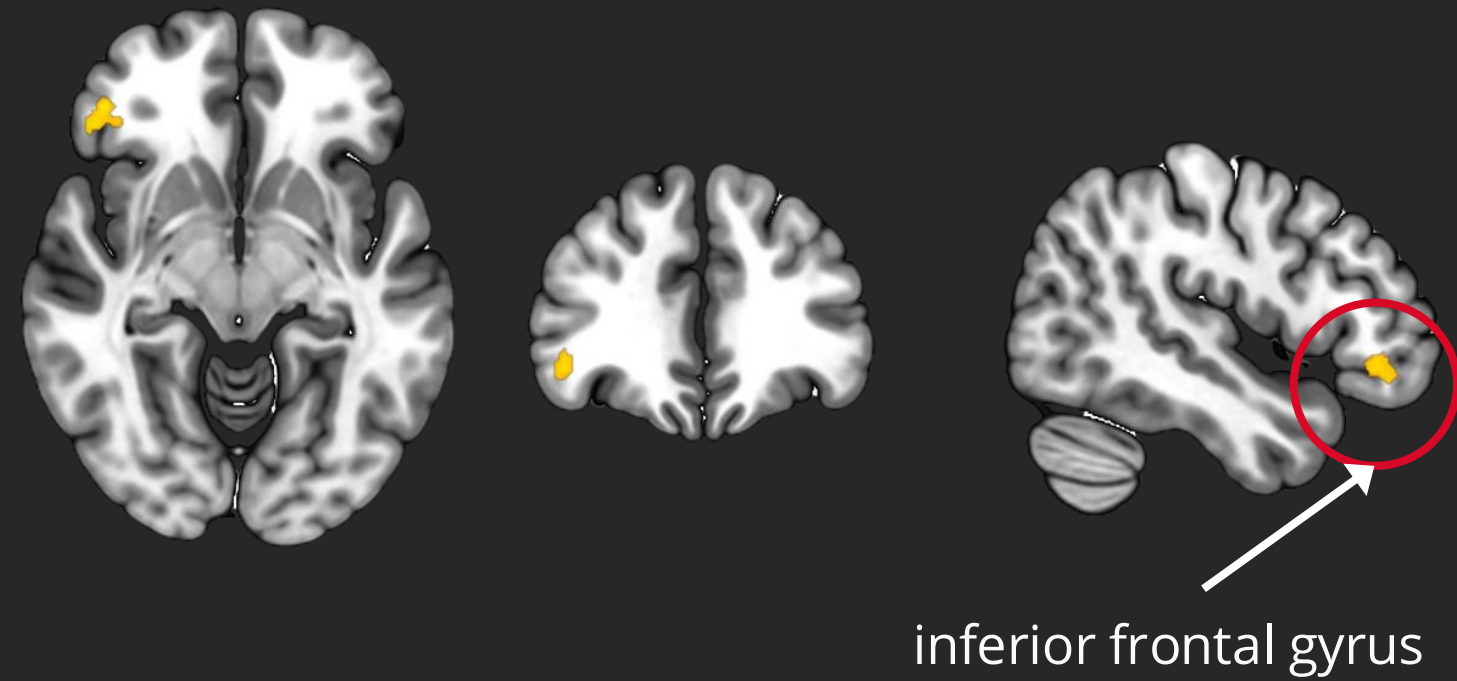


Internalising symptoms:

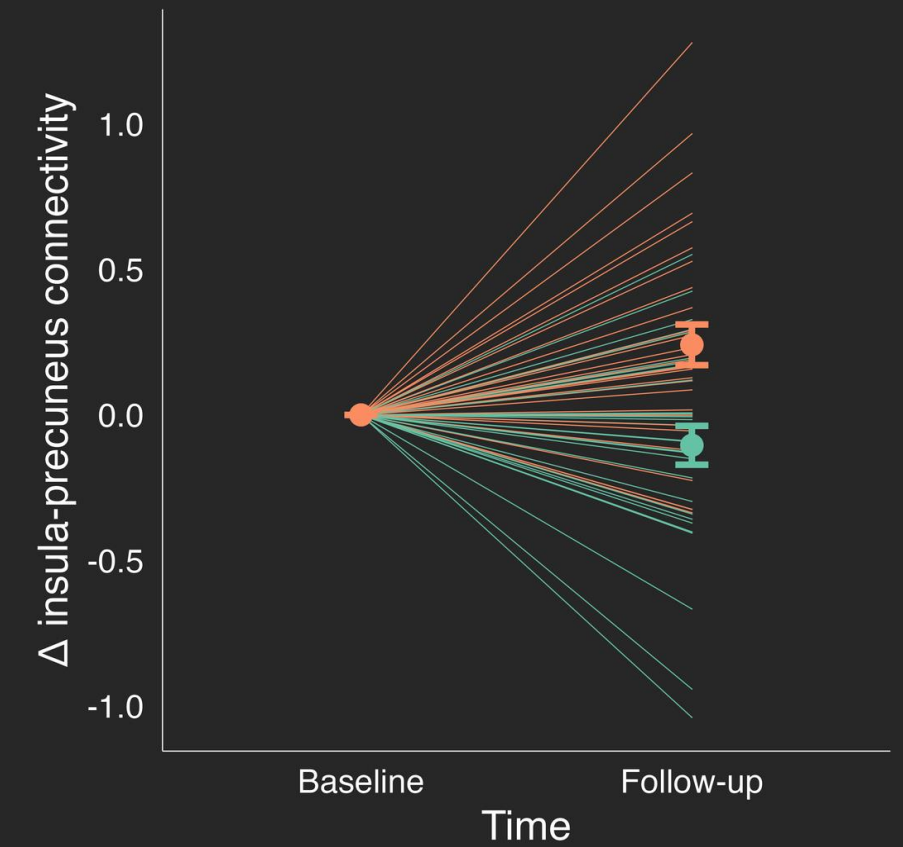
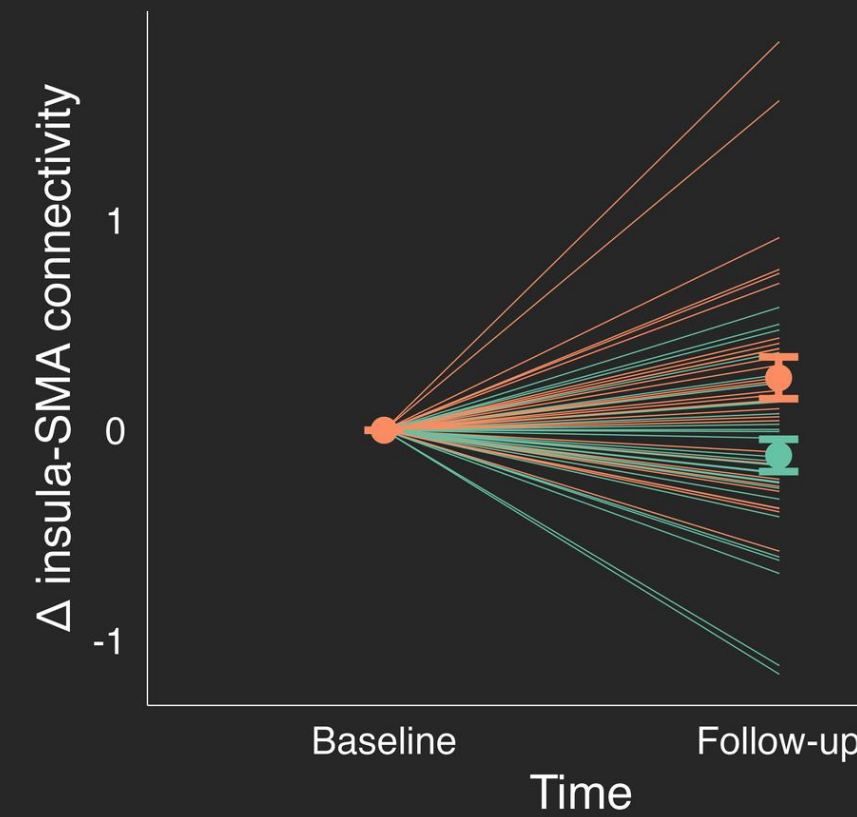
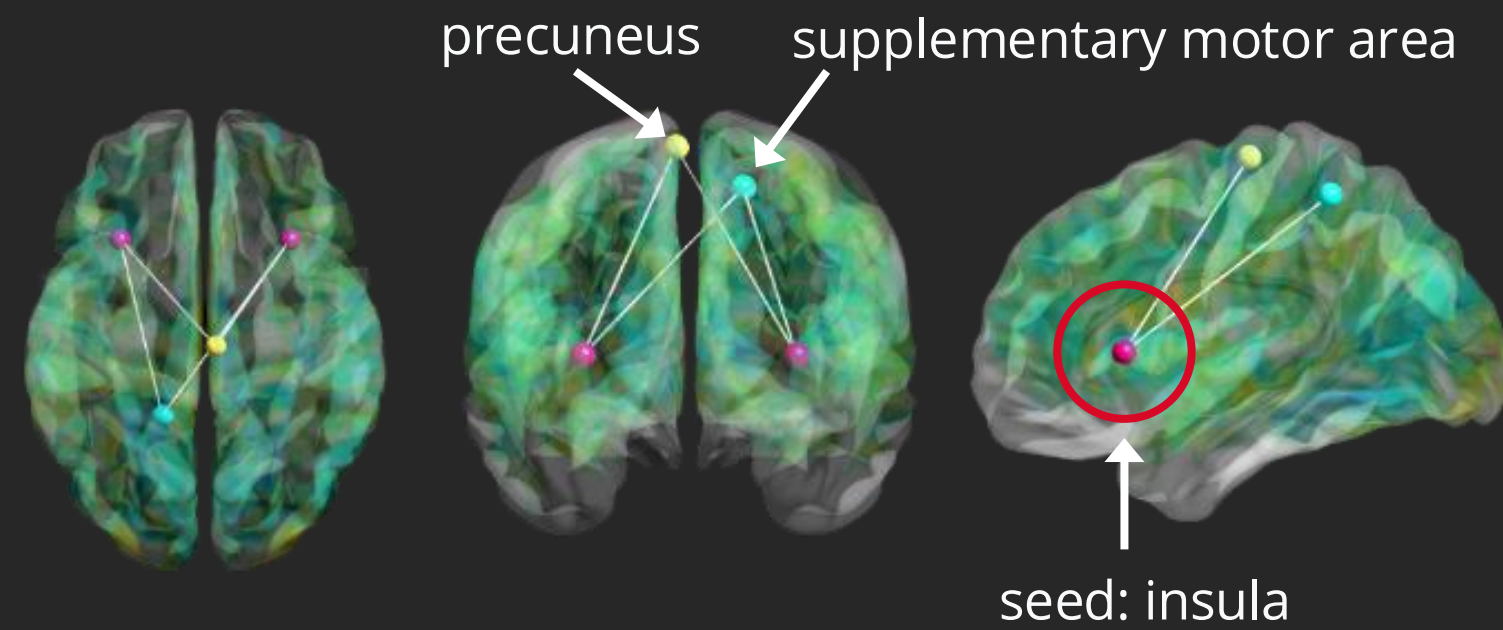
Revised Child Anxiety and Depression Scales
(RCADS) - adolescent & mother reports



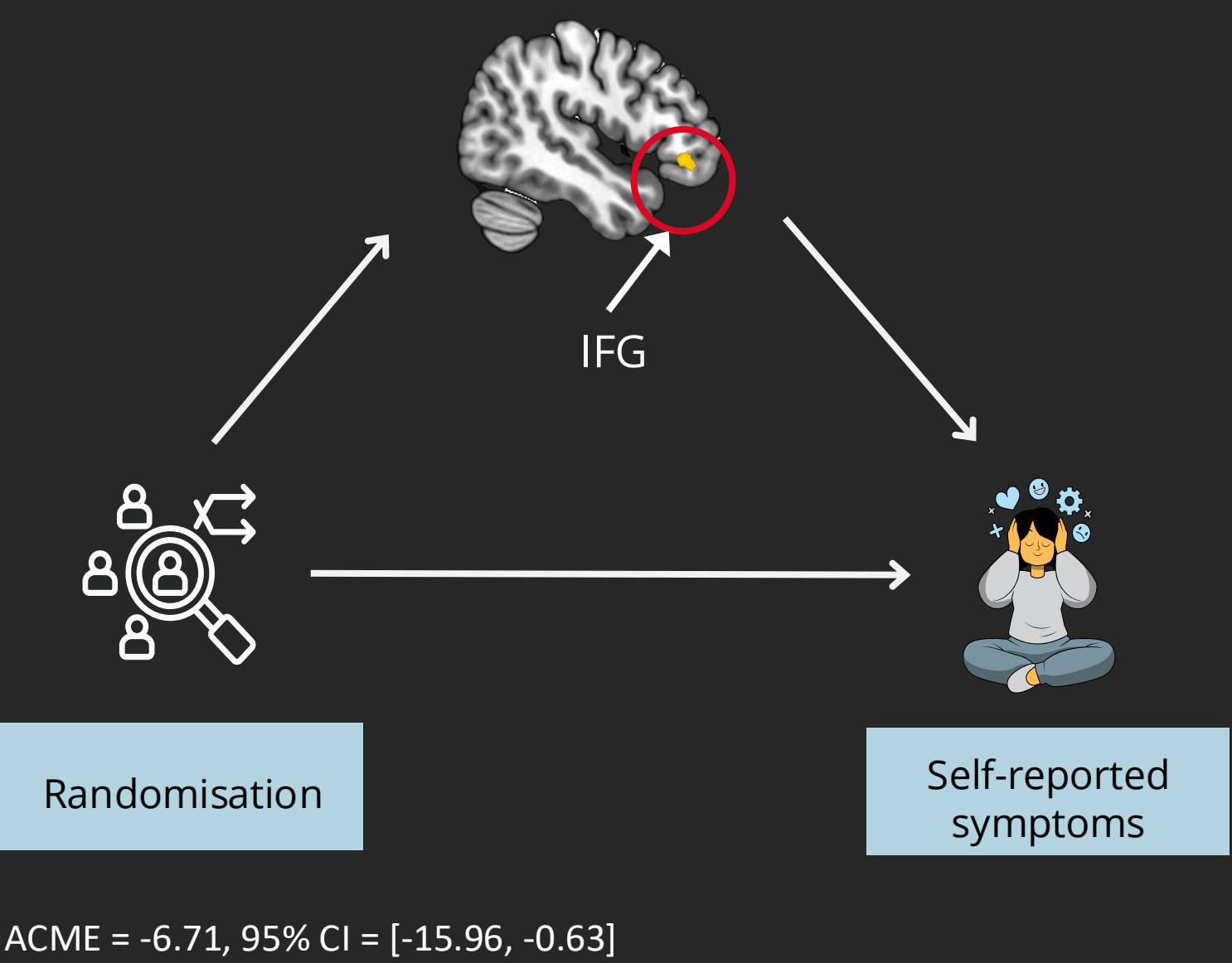
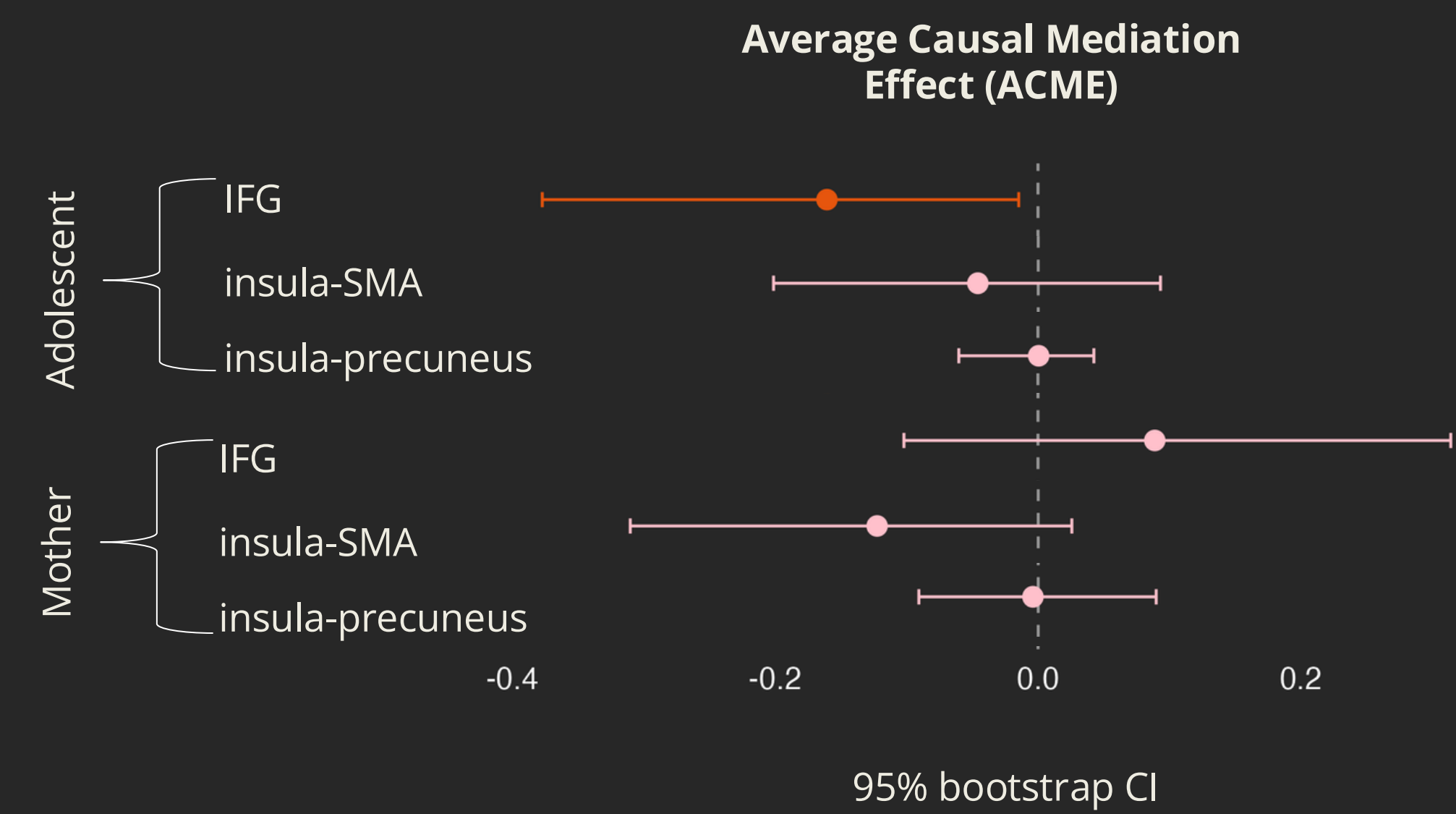
Intervention effects on brain activation



Intervention effects on functional connectivity



Intervention effects on mental health was mediated by change in IFG activity





Causal models needed

Adolescent voices matters

Interventions for parents of adolescents



1 out of 17

parenting intervention studies
for parents of adolescents

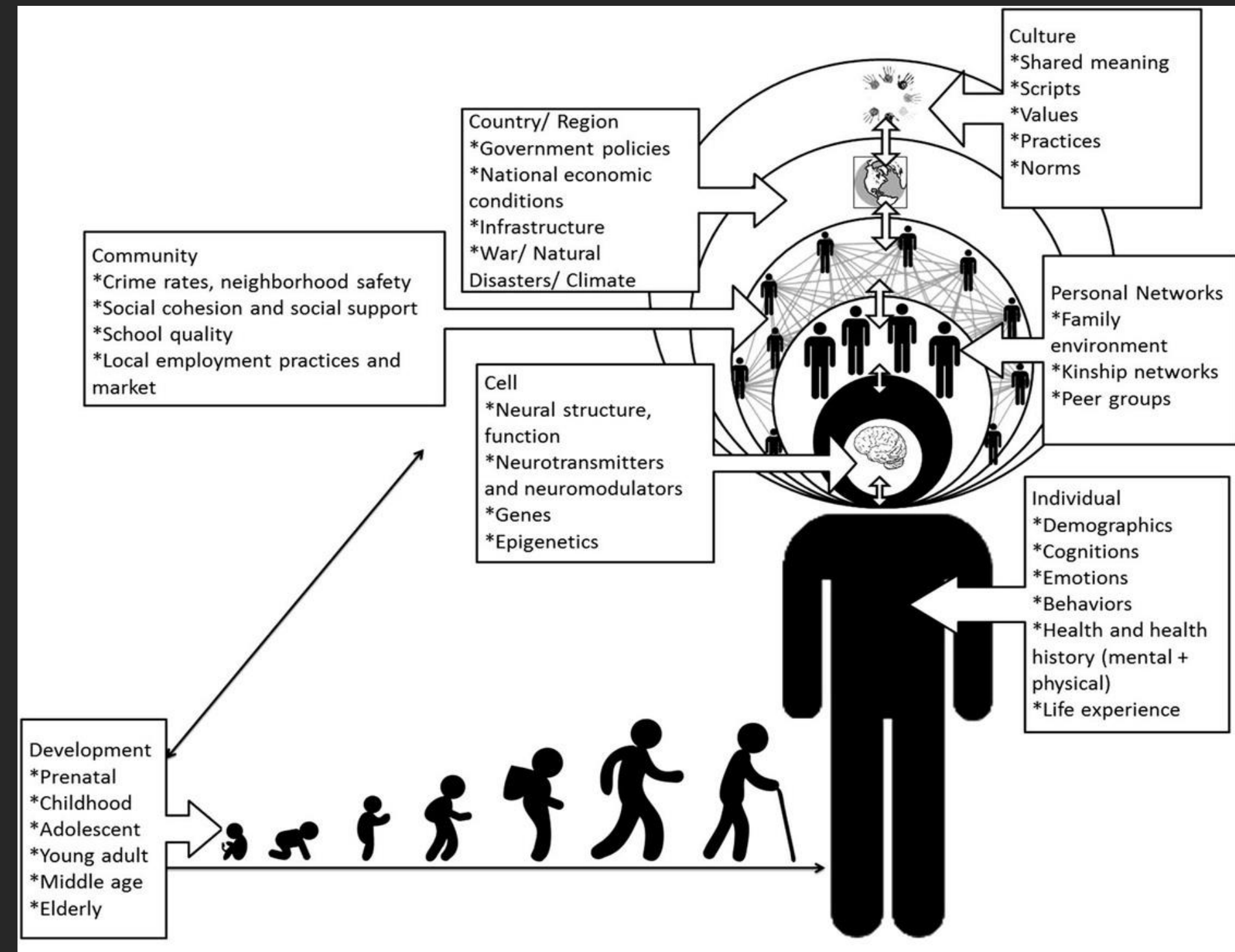
“

Given that families and parents remain the most important figures in the lives of most adolescents, the paucity of rigorous research into family influences on adolescent health and wellbeing is a striking knowledge gap.

”

The Lancet Commission on adolescent health and wellbeing, 2016

So, are we blaming parents?



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Thank
you

"I've learned that people will forget
what you said, people will forget what
you did, but people will never forget
how you made them feel."

Sarah Whittle
Elena Pozzi
Christiane Kehoe
Junxuan Zhao
Sophie Havighurst



• All participants
and their
families!